Replication Package README

Overview

This replication package accompanies the article:

Wildfire Risk and Power Network Infrastructure in Quebec: Climate Projections and Asset Replacement Cost Estimation

Authors: Frédéric Pitz, Charles-Antoine Gosselin, Pierre-Olivier Pineau, Joël Cuerrier, and Hicham Dakkak

Affiliation: Energy Sector Management Chair, HEC Montréal

Contact: frederic.pitz@hec.ca

This paper develops a spatially explicit probabilistic framework to estimate the annual expected replacement cost of overhead power network assets operated by Hydro-Québec under projected wildfire risk through the end of the 21st century. Fire cycle projections from three global climate models (CanESM2, HadGEM2-ES, MIROC-ESM-CHEM) under two emission scenarios (RCP 4.5 and RCP 8.5) are combined with a high-resolution fuel combustibility raster (MRNF) to derive asset-level annual fire probabilities. Probabilistic cost estimates are calibrated against observed historical fire perimeters using a delta scaling approach. Three infrastructure growth scenarios (stable, business-as-usual, energy-efficiency) are evaluated across three 30-year horizons (2011–2040, 2041–2070, 2071–2100).

Project Structure

```
Replication_package/
 Code/
    00_Main.R
    01_Prob_ff.R
    02_Calibrated_costs.R
    03_Stats.R
 Data/
   (see Data section -- must be populated manually)
 Sortie/
     01_prob_ff/
        probabilites/
        redistribution/
     02_calibrated_costs/
        couts/
        historique/
```

```
    calibration/
  03_stats/
```

The `Data/` folder must be populated by the user from the sources listed in the Data section below. The `Sortie/` folder is initially empty and is fully generated upon script execution.

Script Descriptions

`00_Main.R` — Installs and loads all required R packages, defines all directory paths, and creates output subdirectories. This is the only file the user needs to launch manually; all other scripts source it automatically.

`01_Prob_ff.R` — Computes annual fire probabilities per Boulanger grid cell ($P = 1/firecycle$) for each model, scenario, and time horizon, then redistributes them to the asset level using the MRNF fuel combustibility raster (score 0–5). Point assets and distribution lines (discretized at 25 m) are joined to their adjusted probabilities for the stable, BAU, and EE inventory scenarios. Options at the top of the script allow the user to select which models and inventory scenarios to process. Outputs are saved to `Sortie/01_prob_ff/`.

`02_Calibrated_costs.R` — Computes raw expected annual replacement costs from the adjusted fire probabilities and unit replacement costs (Table 1 in the article), estimates realized historical costs (1981–2010) from observed fire perimeters to derive asset-specific delta calibration factors, and applies the correction to all prospective estimates. Options at the top of the script allow the user to select which models and inventory scenarios to process. Outputs are saved to `Sortie/02_calibrated_costs/`.

`03_Stats.R` — Computes descriptive statistics for all 19 Boulanger fire cycle rasters (mean, median, percentiles, share of cells below 100 and 500 years) at the province level and by administrative region. Outputs are saved to `Sortie/03_stats/`.

Outputs

All figures and tables from the article are generated automatically and saved in `Sortie/`. The main outputs are listed below.

File	Description	Location
s2_1_series_par_inventaire_calibre	Figure 4 — Annual expected replacement cost by inventory scenario and emission pathway	02_calibrated_costs/ calibration/graphiques/
s2_3_bar_actifs_calibre_ENG	Figure 5 — Average annual cost by asset type (RCP 4.5 and RCP 8.5)	02_calibrated_costs/ calibration/graphiques/
s3_1_regions_bar_calibre_ENG	Figure 6 — Average annual cost by administrative region	02_calibrated_costs/ calibration/graphiques/
histogram_comparison_RCP85_2071_QC	Figure 3 — Fire cycle duration distribution: baseline vs. RCP 8.5 (2071–2100)	03_stats/
table1_stats_globales.csv	Province-wide fire cycle statistics by GCM × scenario × horizon	03_stats/
table2_stats_par_region.csv	Fire cycle statistics by administrative region	03_stats/

Data: Availability and Provenance

No data are included in this replication package. All datasets must be downloaded independently by the user from the sources listed below. All data are publicly available, with the exception of the Hydro-Québec infrastructure asset inventories.

Dataset	Source	Coverage
Fire cycle projections	Boulanger (2024), Zenodo doi:10.5281/zenodo.12811118	Canada, 0.25°, 1981–2100
Fuel combustibility raster	MRNF (2024), Direction de la protection des forêts donneesquebec.ca	Quebec, 25 m × 25 m
Historical fire perimeters (FEUX_PROV)	MRNF (2024), Direction des inventaires forestiers donneesquebec.ca	Quebec, since 1976
Administrative boundaries (regio_s.shp)	MRNF, donneesquebec.ca	Quebec administrative regions
Asset inventories — stable	Hydro-Québec (restrictions apply; requests to be addressed directly to Hydro-Québec)	Full HQ network
Asset inventories — BAU and EE	Pineau et al. (working paper, HEC Montréal, 2025)	Quebec, 2022–2100
Unit costs — distribution assets	Fant et al. (2020)	U.S. (applied to Quebec)
Unit costs — transmission poles	MISO (2024), MTEP24 guide cdn.misoenergy.org	USD 2023, converted to CAD
Unit costs — distribution transformers	Régie de l'énergie du Québec (2021) hydroquebec.com	CAD, reference year 2021

Computational Requirements

The analysis was conducted using **R version 4.5** on Windows. All required CRAN packages are installed automatically when `00_Main.R` is executed. Key packages include: `terra`, `sf`, `arrow`, `dplyr`, `tidyr`, `ggplot2`, `patchwork`, `scales`, `readr`, `purrr`, `fixest`, `modelsummary`, and `here`.

Replication Instructions

1. Download the `Code/` and `Sortie/` folders from this package.
2. Create an R project at the root of the replication package directory.
3. Download all required datasets (see Data section above) and place them in the `Data/` folder, respecting the subfolder structure expected by the scripts.
4. Update the file paths in the main script and at the top of each script to match the location of each asset inventory file on your system. Users who only wish to replicate the stable inventory results do not need the BAU and EE inventory files.
5. Execute `Code/00_Main.R`. This will install all required packages, initialize paths, and run all scripts sequentially.

All figures and tables are generated automatically and saved in the `Sortie/` directory.

License and Notes

Users wishing to cite this replication package may use the following reference:

Pitz, F., Gosselin, C.-A., Pineau, P.-O., Cuerrier, J., & Dakkak, H. (2026). *Replication Package for "Wildfire Risk and Power Network Infrastructure in Quebec: Climate Projections and Asset Replacement Cost Estimation"*. Energy Sector Management Chair, HEC Montréal.

This README follows best practices for replication package documentation as outlined by Vilhuber et al. (2022).